

Homework 1

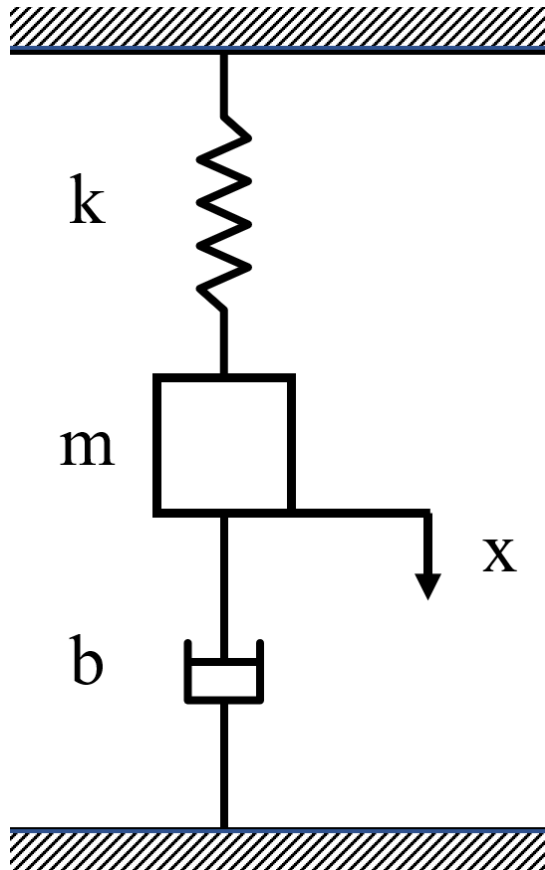
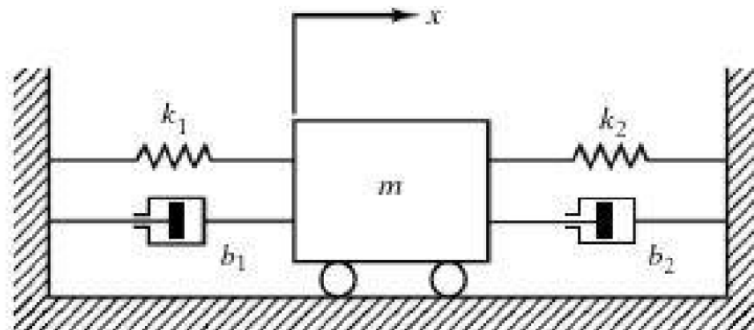
Jizhou Yan ^{TA}

Instructions before the assignment 作业前须知:

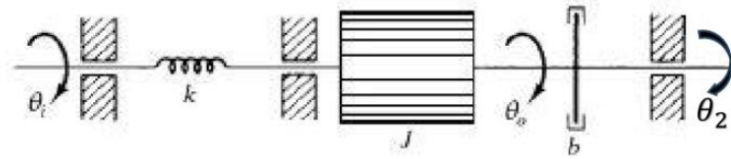
- This assignment is for the **first week's homework** of the System Dynamics and Control course (hereafter referred to as SDC). The submission deadline is **before 11:59 PM on September 19, 2024**. Please ensure timely submission, as late submissions will result in point deductions. 本次作业为 SDC 第一周课程作业, 提交时间为 **2024/09/19 23:59** 前。
 - If submitted within 12 hours after the deadline, no points will be deducted (**this is allowed only twice, so focus on completing it properly rather than rushing to meet the deadline if you're unable to finish on time**);
 - If submitted within three days after the deadline, **20% of the total score of this assignment will be deducted**;
 - If submitted within two week after the deadline, **30% of the total score of this assignment will be deducted**;
 - Any other late submissions will result in a **40% deduction**;
- For late submissions, please contact the teaching assistant(TA), **Yan Jizhou**, via email at **yanjizhou312@163.com** or WeChat **y3139838452**. Make sure to use the filename format as "SDC-Assignment-x-Resubmission-Name-Student ID." 作业补交时通过邮箱或微信与助教闫济洲联系, 邮件标题及文件名备注为 "SDC-第 x 次作业补交-姓名-学号".
- The assignment is graded out of 10 points. It consists of basic course questions (10 points) and extension questions (0 to 5 points). **The final assignment score for each submission will be the smaller value between 10 and the actual score.** Each assignment earns all points by completing only the basic assignment. 每次作业仅完成基础作业即可获得所有分数, 鼓励同学们完成拓展课程题目。
- Students are encouraged to attempt the extension questions. Any extra points earned will be proportionally added to the final course grade for possible grade improvement.
- **Academic misconduct such as plagiarism is strictly prohibited in course work. Plagiarism and other academic misconduct will be dealt with in accordance with the relevant regulations of the University.** If you have discussed the assignment with classmates or received help, please note at the beginning of the assignment: "This assignment was completed after discussion with XXX (classmate's name), and part of the thought process was referenced from XXX (classmate's name)." **Every contribution deserves recognition. Unacknowledged similarities between assignments will be treated as plagiarism.** 课程作业中严禁抄袭等学术不端行为。如作业完成过程中经历了同学讨论/同学帮助的情况, 请在作业最前面注明 "本次作业经过与 XXX (同学姓名) 讨论, 参考了 XXX (同学姓名) 的部分思路"。每位同学的帮助都值得被记录。不被注明的作业雷同将被认为是抄袭行为。

1. Problem 1

Derive the equation of motion (EOM) in the following figure. Note that the external force/external moment applied in this question acts on the mass and in the direction of the arrow.

1.1 a**1.2 b**

1.3 c



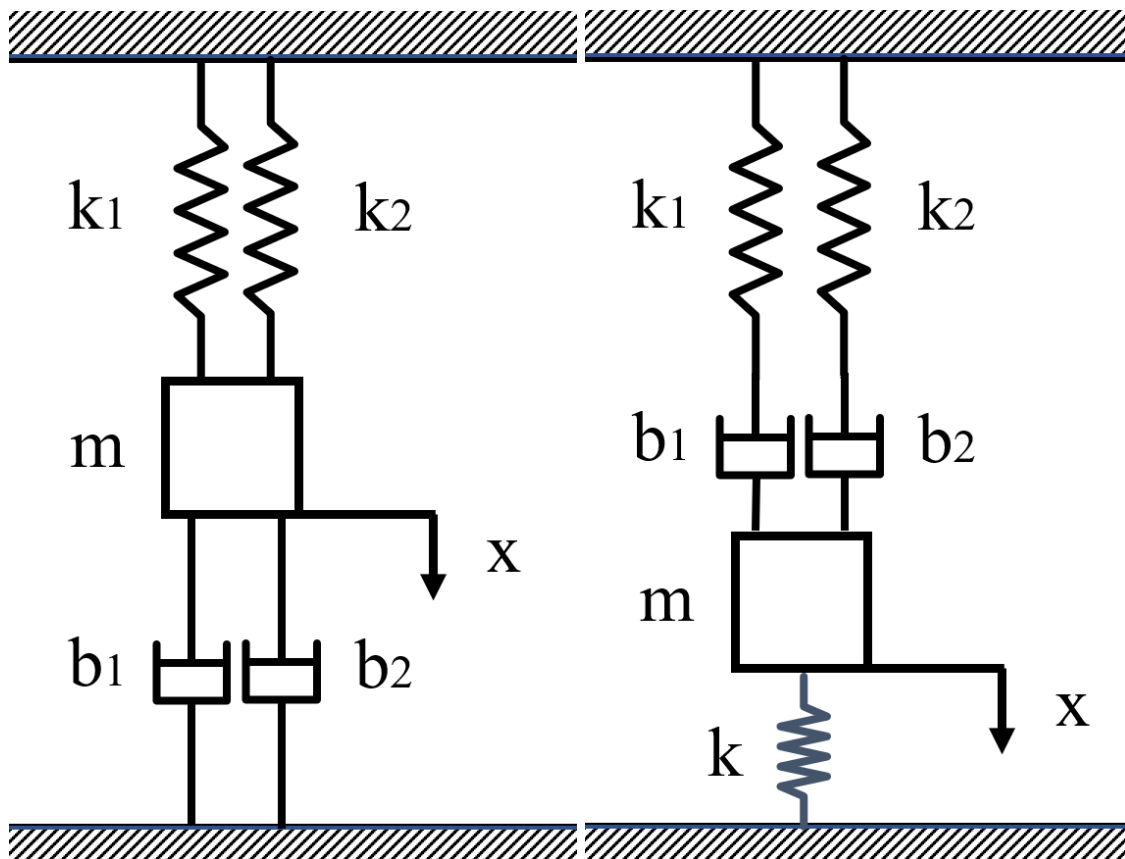
Angle in the figure are $\theta_0, \theta_1, \theta_2$. Derive the EOM for θ_0 .

2. Problem 2

In this problem, we will explore the dynamics equations for a more complex system. Please independently account for any possible external forces that may arise.

2.1 a

Derive the equation represented by each figure below



2.2 b (Dynamics equations expressed in vectors)

In multi-degree-of-freedom systems, we often use a state vector to represent the system's state. Here, we provide an example where the coordinates in 3D space are used as the state vector.

Assume that there is a tetrahedron with a particle of mass m located at the center of the tetrahedron. Each of the vertices of the tetrahedron is connected to the particle by a shock absorber, which can be approximated

as a spring and a damper, with parameters k and b . Please derive the dynamics equation for this system, and express the resulting dynamics equation in matrix form.

If the particle is not at the center of the tetrahedron but at a position displaced by a vector $\mathbf{r}$ from the center, please derive the dynamics equation for this case.

3. Problem 3

In this problem, we will explore the dynamics equations of a particle system.

Although referred to as a particle system, this is essentially just another way of increasing the degrees of freedom. In this problem, we recommend using **generalized coordinates** to represent the system's state. We also expect students to convert the final dynamics equations into matrix form.

3.1 Tips

Using the Lagrange method can simplify the calculation.

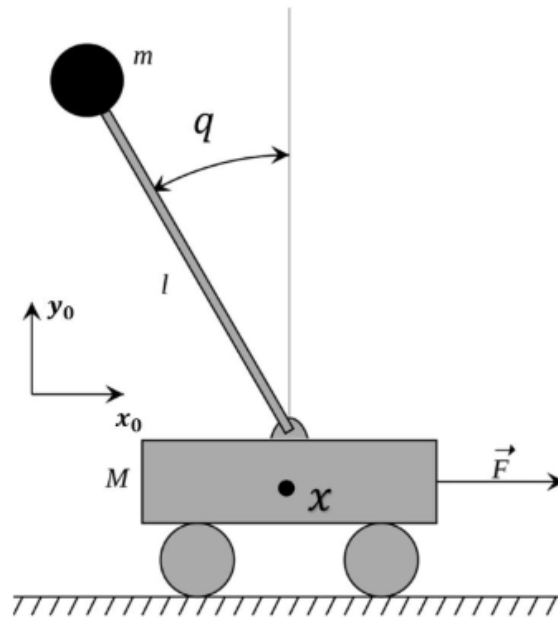
The Lagrange function of the system is

$$L = T - V$$

where T represents the kinetic energy and V represents the potential energy. Finally, the EOM can be derived from the Lagrange equation

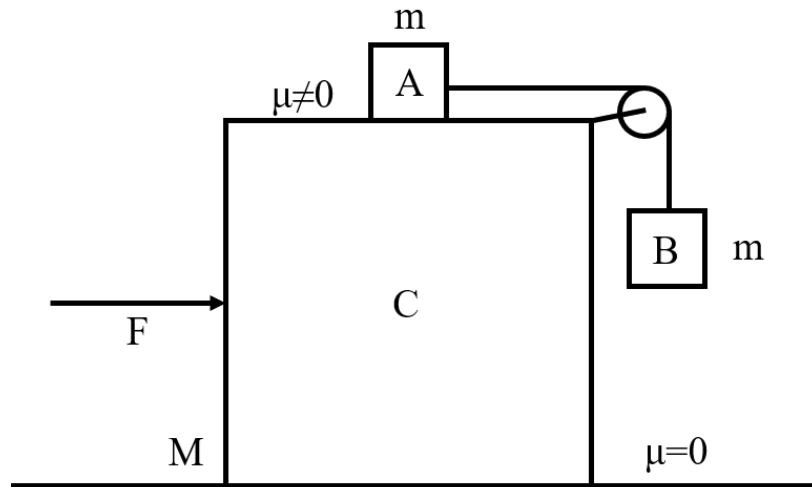
$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = Q_i$$

3.2 a



In the diagram, we have a first-order inverted pendulum system. Neglect the mass of the rod and all forms of friction, and derive the dynamics equation when the pendulum deviates by a **small angle**. (Note: The centrifugal force term will lead to nonlinearity in the system equation. If you wish to simplify the problem to a linear one, please ignore the centrifugal force term.)

3.3 b



In the figure, there are three particles, A, B, and C. Please design the state variables for the system and derive the system's dynamics equations. (Note: particle B may experience oscillations due to acceleration.)

4. *Problem 4 (breadth extension)

Beyond the field of dynamics, many other areas also involve system modeling and control. In this problem, we will explore system modeling in the context of biological communities. Please supplement any additional parameters as needed.

4.1 a

In a vast Qingqing grassland, there are two different biotic populations, referred to as "Xi" and "Hui". The interspecies relationship between these two organisms includes both **predation** and **mutual cooperation** (since they cooperate at least once a year :yum:). The population sizes of the two groups are denoted by x and h , and their relationship is governed by the following:

1. The growth rate of each population is proportional to its own population size, with proportionality constants a and b , respectively.
2. The growth rate of x is linearly related to h with a slope of $-c$.
3. The growth rate of h is linearly related to x with a slope of $-d$.

Here, a, b, c, d are constants.

Please derive and solve this system of equations. Also, indicate the conditions under which this system might collapse. (You cannot expect a population size to be negative or have infinite organisms on the Qingqing grassland.)

4.2 b

It can be observed that in the previous question, we only considered the predation relationship. We will now further consider the mutualistic relationship. In this relationship, the rate of change of the growth rates of the two populations is linearly related to the sum of the population sizes, with a slope of $-e$.

This means that as the sum of the populations decreases, the growth rate of each population increases, while as the sum of the populations increases, the growth rate of each population decreases.

Derive the equations for this system. You don't have to solve this equation.